Introduction to Health Economics

Gaurav Dhamija

Indian Institute of Technology Hyderabad

Overview

- Objective: Build an understanding of economics of the health and health care delivery.
- List of Topics
  - Demand for health and health care delivery
  - Supply of health care
  - Asymmetric information in the Health Insurance Market

Overview

- Evaluation Procedure:
  - Two Quizzes: 60%
  - Final Exam: 40%
- Office Hours: Tuesday and Thursday (4:00 to 5:30 pm)
- Reference Material
  - Bhattacharya, J., Hyde, T., Tu, P. (2013). Health economics. Macmillan International Higher Education. – Mandatory Text

Introduction to Health Economics: Motivation

- Economics is defined as the science that studies human behavior in relation to ends and scarce means with alternative uses, focusing on decision-making, efficiency, and the identification of feasible alternatives.
- Why do economists work in health?
  - Should it not be about medicine, nursing, caring and the difference between life and death?
  - It is not supposed to be about money, profit, production and markets.
- Such views demonstrate a limited understanding of the role and content of economics.
- In principle, economists are concerned with better choices and in particular making the best use of existing resources and growth in the availability of resources.

- Moreover, health is different from other goods, and a source of “special economic problems”, for one major reason: uncertainty (Kenneth Arrow).
  - Demand for other goods such as bananas or televisions is predictable. On the contrary, demand for health care is highly uncertain.
  - An unforeseen broken leg or epidemic can suddenly create demand for expensive health care services.
  - Risk averse individuals find this health-related uncertainty as unpleasant. We will see that property of risk averseness derives the demand for health insurance.

- Think of the supply side of the insurance
  - A man walks into the office of a life insurance company.
  - He wants to buy a INR 1 million life insurance policy for a term of one day. Your company will have to pay INR 1 million to his heirs if and only if he dies tomorrow.
  - You know nothing else about this man.
  - Would like to do sell this policy of this person? What is the problem?
  - Asymmetric information: a situation in which agents in a potential economic transaction do not have the same information about the quality of the good being transacted
- We will see that it is the source of many problems such as 'Adverse Selection' in health insurance markets and it can break down the market also.

- Asymmetric information can also lead to moral hazard:
  - An individual faces some risk of a bad event X, and his actions can increase or decrease its likelihood.
  - The individual buys an insurance contract that will help pay some or all of the costs of X if it occurs. The price of X to the individual is now lower – this price distortion means the individual does not face all the consequences of his actions.
  - In response to the price distortion, the individual changes his behavior in a way that increases the chances of X or increases the costs of recovering from X.
  - The insurance company cannot observe this behavior change, because there is an information asymmetry. Otherwise the contract would have been written to discourage or penalize the riskier behavior
  - The individual's riskier behavior creates a social loss, because the costly event X occurs more than it would have without insurance.

- Health care markets are rife with externalities because health status is a uniquely contagious quantity.
  - It probably does not matter very much to you if your roommate decides to eat a banana.
  - However, it certainly does matter if your roommate decides to smoke in the common place.

- Are people actually price-insensitive when it comes to health care?
- Or does demand for health care respond to price, even for health care that may be a matter of life and death?

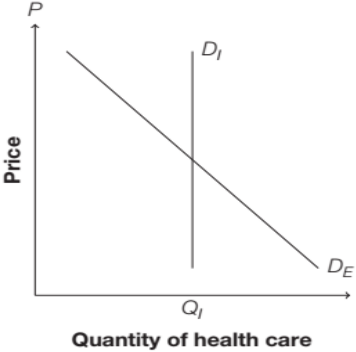
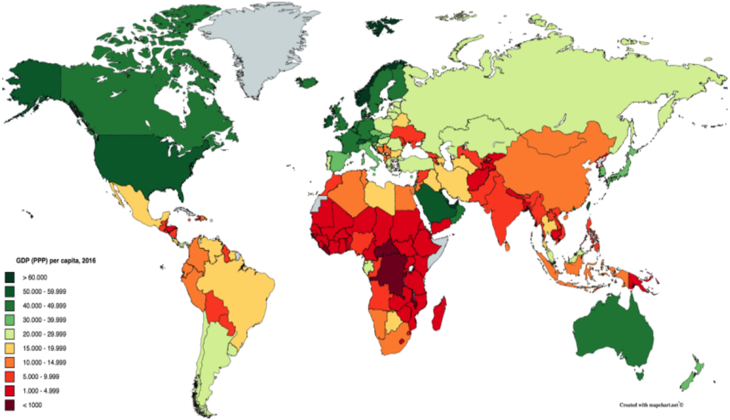


Figure 1: A price-inelastic demand curve, D_I, and price-elastic one, D_E

- Figure 1 lies at the center of health economics.

- Another reason why economists are fascinated towards health outcomes and health care is its relationship with economic development.



Source: Pascaline Dupas (WS 2020)

Figure 2: GDP (PPP) per capita by countries in 2016

- Do you see any relationship between geography and GDP?

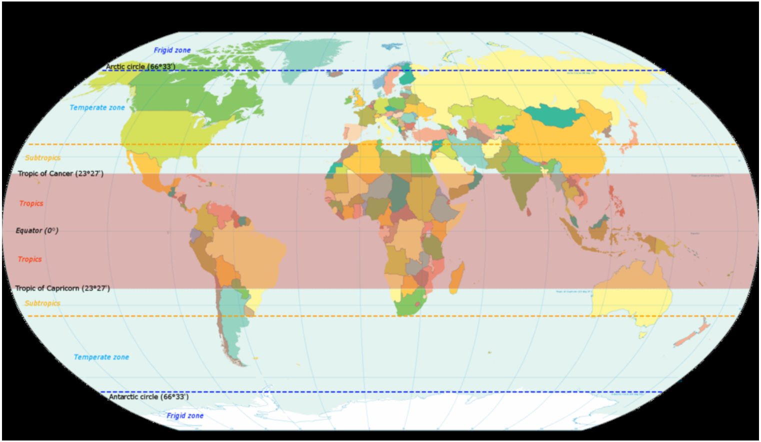
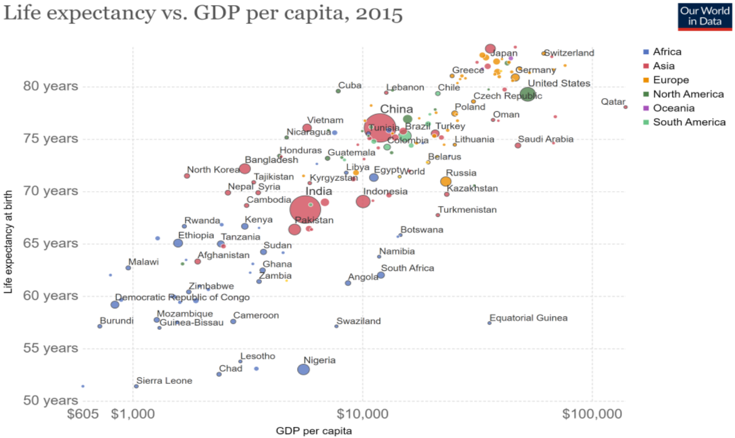


Figure 3: World map indicating tropics and subtropics

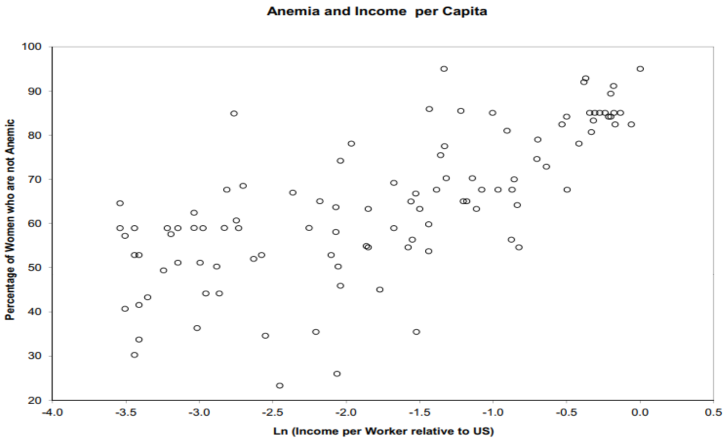
- Tropical regions tend to be poor.
- What is it that makes it harder for the countries in the tropical region to grow faster?

- Hypothesis: Prevalence of tropical diseases such as malaria, yellow fever, dengue, ebola, helminths, sleeping sickness, lymphatic filariasis, schistosomiasis, ... affects development.
- There are several pieces of evidence to support this hypothesis.



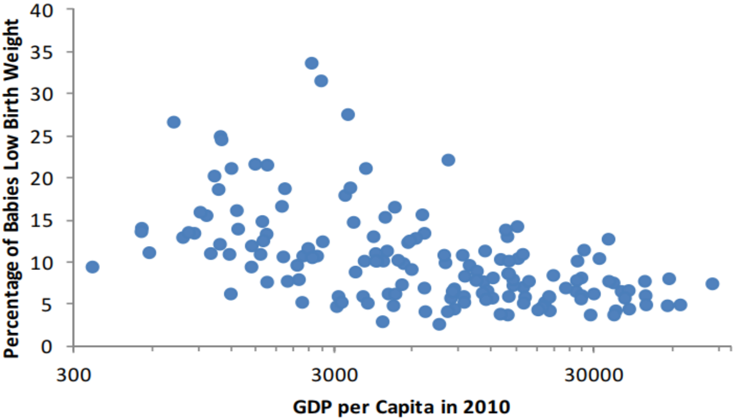
Source: Clio-Infra & UN Population Division; Maddison Project Database (2018)

Figure 4: Correlations between Health and Income – Preston Curve



Source: Shastry & Weil (JEEA 2003): "How Much of Income Variation is Explained by Health?"

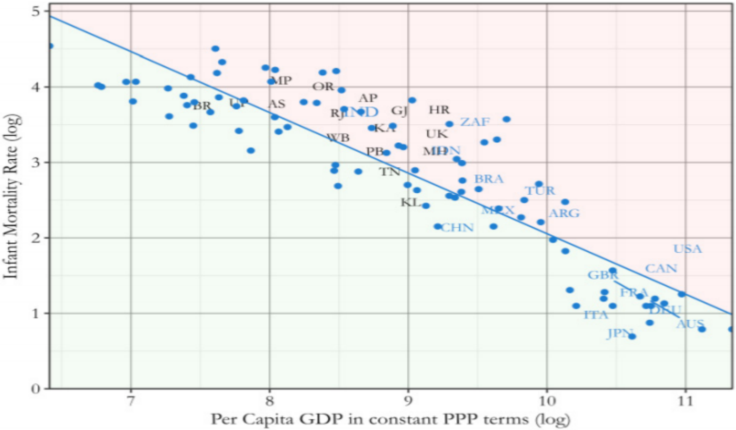
Figure 5: Correlations between Anemia and Income



Source: Pascaline Dupas (WS 2020)

Figure 6: Correlations between LBW and Income

- This cross country relationship holds true within country too.



Source: IGC policy note (2017)

Figure 7: Correlations between IMR and Income

- Which way does the causality go?
  - Health $\Rightarrow$ Income : Health matters for productivity
  - Income $\Rightarrow$ Health : Health is a normal good and rises with income.
- Historical data on health supports the second hypothesis
  - Prior to the Industrial Revolution:
    - there was little long-term change in average health, though with considerable short-run variation (e.g. Black Plague)
    - cross-sectional differences (across countries) in health were small
  - In the 19th century, there was a takeoff of health in Europe and its offshoots, with little change elsewhere
  - Starting in the middle of the 20th century, health improvements in trailing countries began to exceed those in leaders
  - Data suggests that health convergence in last 50 years is much stronger than income convergence

- Historically, improvements in health resulted from three sources:
    - Improved standard of living, especially nutrition
    - Public health investments (which rise with income) – clean water, sanitation
        - Cutler and Miller (2005): 43 percent of decline in mortality in 13 big US cities 1900-1936 due to water quality improvements
    - Medical innnovations (antibiotics, other drugs, surgery, etc.)
- In high-income countries, these happened roughly in the order listed.
- However in catch-up countries, the order was different.
    - benefited from medical discoveries in high-income countries, so arguably "exogenous" shock to health (shifting up of Preston Curve)

- Shifting of Preston Curve.

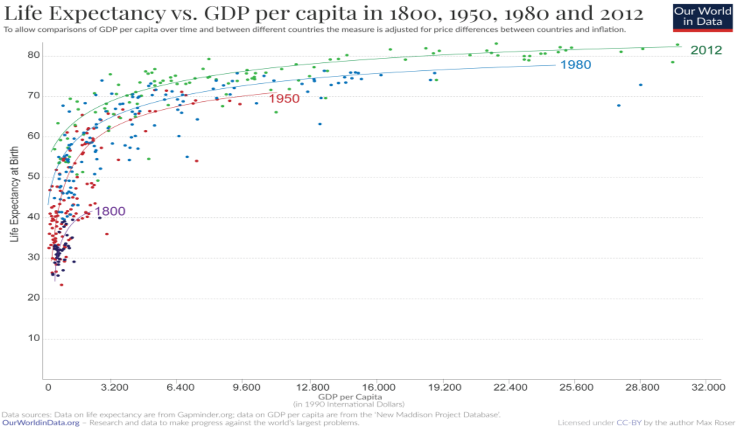


Figure 8: Correlations between life expectancy and Income

- However, the first hypothesis (Health $\Rightarrow$ Income) cannot be ignored given the effect of pandemic on economic growth across the countries.

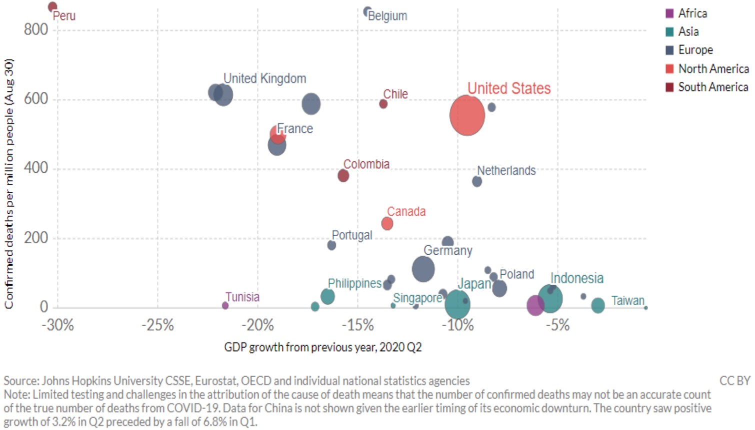


Figure 9: Correlations between COVID-19 deaths and Income

- Gallup and Sachs (2001) quantified the association between malaria and the level and growth of per capita income
  - period 1965-1995
  - cross-country regression framework controlling for historical, geographical, social, economic, and institutional country characteristics
  - malaria endemic countries had, ceteris paribus, per capita income levels 70 percent lower than those of nonendemic countries
  - a 10 percent reduction in the malaria exposure index was associated with a 0.26 percentage point increase in annual per capita growth rates.
- Revisited by Sarma et al.(2019)
  - period 2000-2017
  - exploiting Roll Back Malaria efforts to estimate country fixed-effects model
  - a 10 percent decrease in malaria incidence is associated with an increase in income per capita of nearly 0.3 percent on average and a 0.11 percentage point faster per capita growth per annum.

- What are the possible mechanisms that explain this hypothesis?
  - Direct effect of disease on productivity
    - Disease burden ⇒ Poor nutrition ⇒ Lower productivity
    - Disease burden ⇒ Many illness spells ⇒ Fewer days worked (because sick oneself or caring for sick relative)
    - Disease burden ⇒ Exposure in utero or childhood ⇒ Reduced endowments (physical, IQ) ⇒ Lower productivity
  - Indirect effect of disease on productivity (via lower levels of investments in education
    - Disease burden ⇒ High health care costs ⇒ Reduces income available for investments in education, production, etc
    - Disease burden ⇒ Shorter life expectancy ⇒ Reduced returns to investing in education
    - Disease burden ⇒ High infant mortality ⇒ Higher fertility ⇒ quality-quantity tradeoff
    - Disease burden ⇒ Orphanhood / sick parents ⇒ Need to work as child to compensate (household work or labor market work) ⇒ no time for school
  - Disease burden reduces returns to business and infrastructure investments
    - Disease burden ⇒ Less foreign direct investment ⇒ Fewer resources in-country

- Other potential effects:
  - Disease burden $\Rightarrow$ Outmigration of those who can afford to leave $\Rightarrow$ Fewer resources in-country
  - Disease burden $\Rightarrow$ Less income from tourism
- There could also be indirect effects of disease burden through institutions.
  - Acemoglu, Johnson, Robinson (AER, 2001): "The Colonial Origins of Comparative Development: An Empirical Investigation."
    - Disease burden at the time of colonization determined where europeans settled en masse and where they only sent a few people
    - If no or few European settlers in colonized country, no incentive to establish strong property rights and instead establishment of extractive rent-seeking institutions
    - When there were significant settlements (e.g. US, Canada): Lower strata of European settlers demanded property rights; placed constraints on elites' ability to establish rent-seeking institutions
    - Institutions persisted even after independence
    - So: Disease burden $\Rightarrow$ Potential mortality of European settlers $\Rightarrow$ European settlements $\Rightarrow$ Better early institutions $\Rightarrow$ Better current institutions

- However, there are potential problems with the estimates based on the first hypothesis.
- What we see as the effect of health (disease burden) on development might not be true.
- Tropical disease is a consequence of poverty as well, so reverse causality.
- Hard to isolate effect of health on growth.
- There are estimation techniques which helps in isolating the effects.

References

- Bhattacharya, J., Hyde, T., Tu, P. (2013). Health economics. Macmillan International Higher Education.
- Lecture Slides of Pascaline Dupas (Stanford University) in Winter School 2020.